Leeds University Business School

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Student ID Number:	2	0	2	0	0	7	3	2	7
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Module Leader:	Aritad Choicharoon								
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	Delete as appropriate
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I am aware of the GAI category for this assignment (delete opposite to show the correct one), and have adhered to the guidance for that category.	AMBER

Assignments should be submitted in time but will be accepted (with late penalties) up to 14 days after deadline

Late penalties = 5 marks per 24hours late

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Part 1: Improving Revenue

Introduction

SweetAroma is an e-commerce business that operates as a direct-to- consumer shop for high-end perfume, and as part of the analytics team, this report has been prepared to support business decision-making by means of recent customer order data. One of the main company objectives is to gain knowledge about effects of customer characteristics and online behaviour on spending on the website, in order to improve future marketing activities.

This main goal of this report is to give information about two business requirements set by management:

1. Influence of customer characteristics and on-site behaviour on revenue, using a sample of company transaction data from July 2025 about customer history, website engagement, and promotional activity.
2. Developing a predictive model to estimate revenue, with the aim of identifying potentially high-value customers and supporting certain advertising initiatives.

The performance of the model is evaluated using several metrics, mainly by means of root mean square error (RMSE), then tested on a set of 20 future orsders.

Data Overview and Preparation

The dataset contains information about 10,000 customer orders from July 2025. Every observation here stands for a single order placed on the SweetAroma website and revenue (GBP) is our target feature. Other features are related to customer background and behaviors such as advertising channel, number of past orders, total past spending, time spent on the website, and voucher usage. There is a second dataset regarding 20 new orders and revenue must be predicted using a certain model on that.

From an initial look, only minor percentage of missing values were revealed in all predictor features, ranging from 1.03% to 1.34%. The ratio of missing data to overall dataset size was quite low therefore, rows with missing values could be removed. In the end of the data cleaning section, we left with a clean dataset of 9,419 rows.

Duplicate observations, accounting for approximately 4.84% of the dataset, had to be checked. As they are not data errors, all of them were not touched. There is a high

possibility of customer orders with the same values, especially in an e-commerce context, repeated patterns are quite common. That is why, eliminating such records may cause reduction in the real representativeness of client behavior data.

Regarding revenue outliers, approximately 1.2% of observations turned out to be outside the limits defined by the application of the interquartile range (IQR) method. Nevertheless, considering the possibility of low- and high-spending customers, it was unnecessary to drop those numbers because all were possible scenarios (between 4 and 106). Since the main goal requires to detect orders of high values, excluding these observations could led to wrong results. Thus, the impacts of removing outlier and duplicate values were instead tested separately.

Exploratory Analysis of Revenue Drivers

In this part, the distribution of revenue and its relation with other characteristics were determined. As shown in Figure 1, revenue has an approximately bell-shaped distribution centered around a mean of £55.47, having most values between 47 and 64. Both careful and high spenders can be noticed in this graph, shown by a few low and high order values affecting tails of the distribution. Interestingly, the website browsing time turned out to be the key driver of revenue.

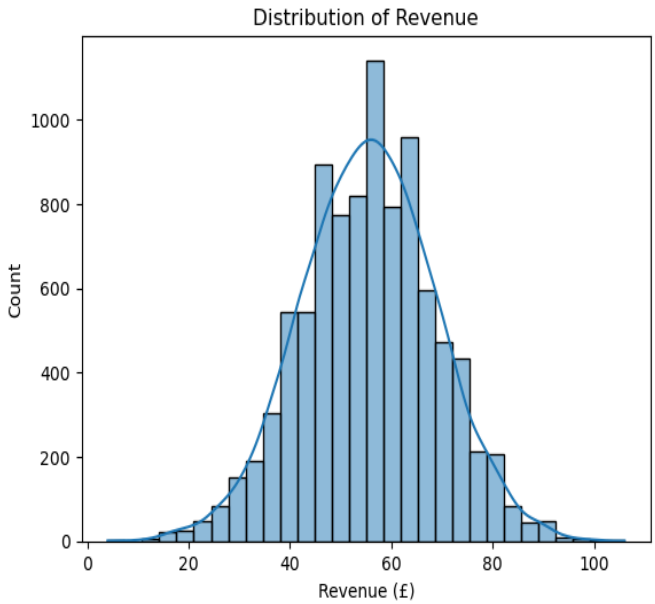

Figure 1.Distribution of order-level revenue (£) with an overlaid density curve.

Figure 2 depicts a strong positive relationship between time spent browsing and order value, meaning higher engagement generally brings to higher revenue. Moreover, the graph (Figure 3) dividing buyers into quartiles based on how much time they spend on the website, is another evidence of this idea. Figures for medians goes steadily up from the lowest to the highest quartile. This means people in the top engagement group are priority clients. Therefore, these findings supports that website engagement has the key influence on customer spending.

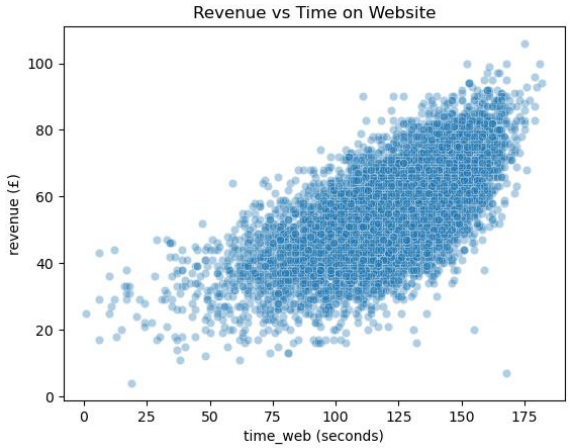

Figure 2.Scatter plot showing the relationship between time spent on the website and order-level revenue (£).

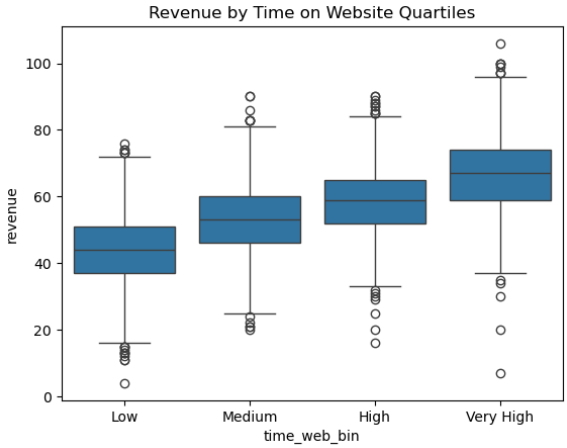

Figure 3.Box plot of order-level revenue (£) across time-on-website quartiles.

On the other hand, weaker direct relationships were found amongst revenue and customer purchase history features which includes past spending and number of past orders. Hence, at the individual order level, there is no dependable rise of revenue with either parameter.

Despite the fact that past spending and number of past orders are significantly correlated with each other (correlation of 0.70), their correlations with revenue are almost zero. This highlights that whilst purchase history

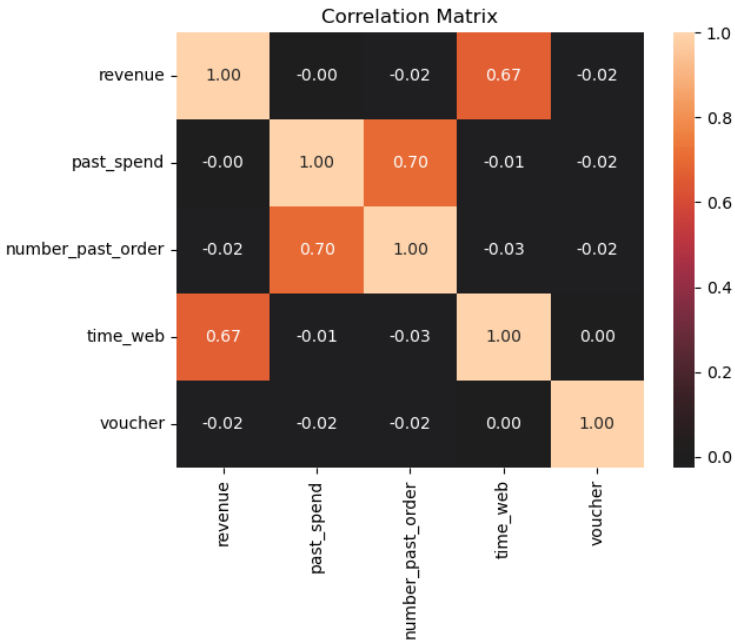

Figure 4. Correlation heatmap of revenue and explanatory variables.

reflects customer behavior in long term, it becomes less useful for forecasting the number for single order (Figure 4).

Furthermore, revenue distributions across advertising channels were also analyzed, the mean and median revenue appeared to be around £55 throughout all channels. This means that the channel through which customers access the website has limited effect on expenditure. Similar pattern was observed for voucher usage which had only a marginal impact on revenue, with customers using vouchers spending negligibly less on average (Figure 5). Overall, it became obvious that engagement-based features are more considerable factors of revenue than customer background or marketing methods. Additional plots are provided in Appendices.

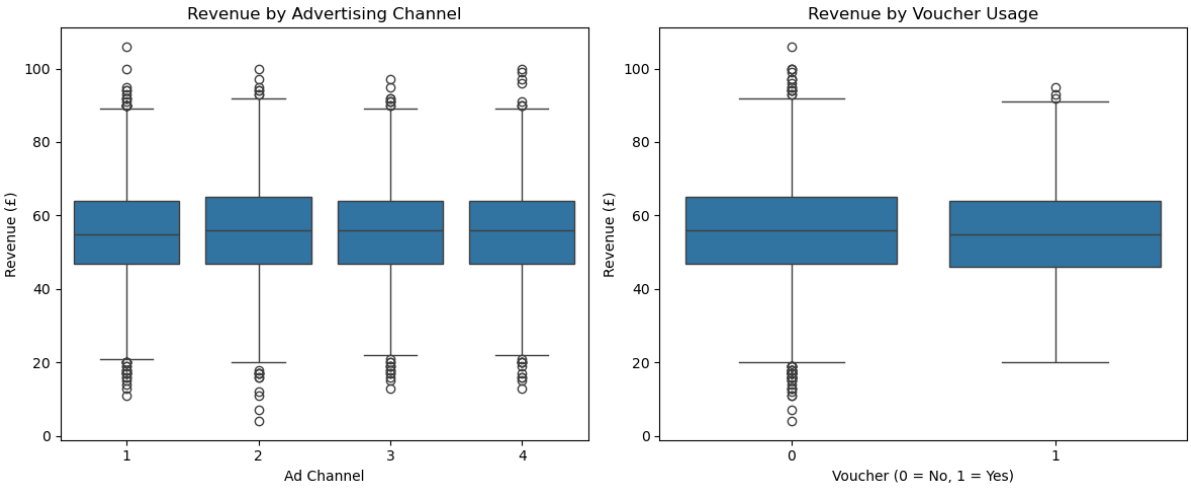

Figure 5. Box plots showing the distribution of order-level revenue (£) by advertising channel and voucher usage.

Model Development

For predicting future revenue, a model called linear regression was created to predict revenue. The reason behind the selection of this model was that our target variable was continuous, and from the exploratory analysis, an approximate linear relationship between revenue and time spent (the main predictor) was detected. The scatter plot

below (Figure 6) was generated with the line slop of 0.35, showing the roughly linear increase in the revenue with browsing period.

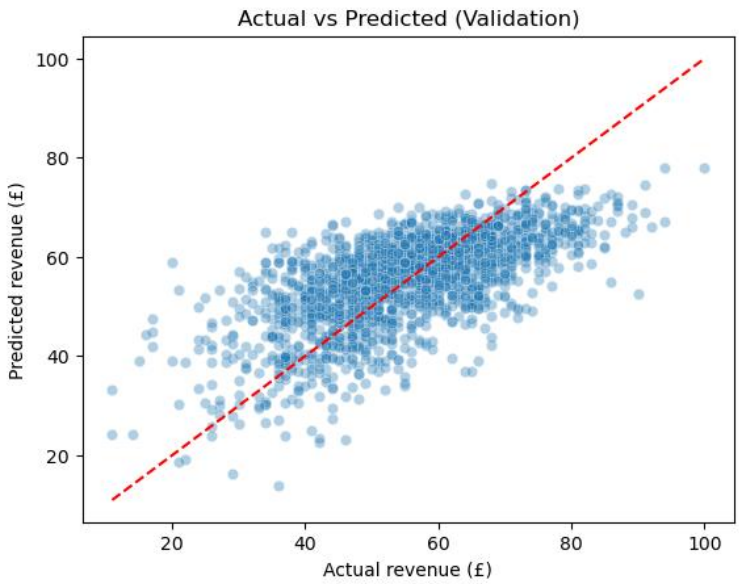

Figure 6. Scatter plot comparing actual and predicted order-level revenue (£) on the validation set, with the dashed line indicating perfect prediction.

While training the model, customer engagement, purchase records, voucher usage, and advertising channels were taken as the predictor features. In this way, the company will able to find out the connection between the changes in the customer behavior related features and income variations. Ultimately, a linear model appeared to be an ideal fit to meet the business demand for transparency and simple understanding.

Model Evaluation and Robustness

The model performance was tested by means of root mean square error (RMSE) as required, using a train–validation split. Another metrics, namely mean absolute error (MAE) and R^2, were also used for further accuracy.

On the validation set, an RMSE, MAE and R^2 of the model were equal to 9.89, 7.83, and 0.42, respectively. These shows that our model can explain an important amount of the variations in the target feature but a single customer behaviour is hard to understand.

Residual analysis demonstrated an approximately symmetric distribution centered around zero to confirm the linear model assumptions made and inexistences of systematic bias. Even though, some deviations can be observed at higher revenues, the model captures the overall pattern in customer spending, which can be seen in a comparison of actual and predicted revenue (Figure 7).

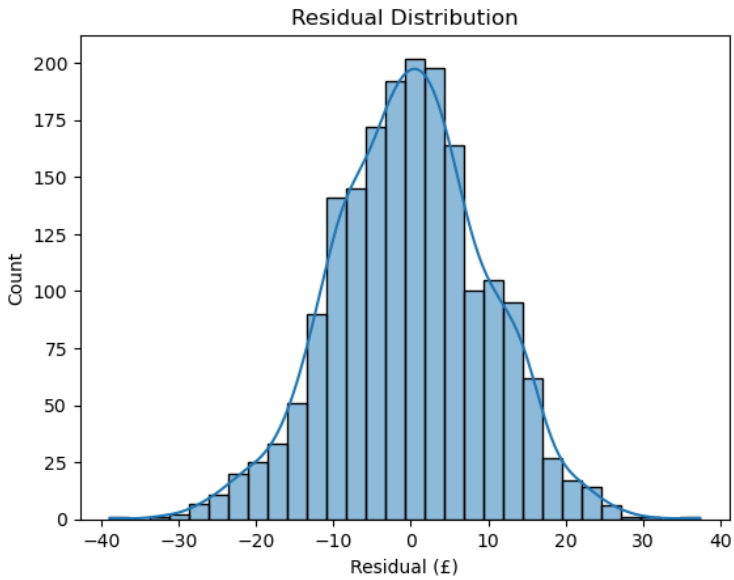

Figure 7. Distribution of model residuals (£) with an overlaid density curve.

With the purpose of assessing the robustness of the modelling assumptions, the model was re-estimated to see If removing duplicate observations and revenue outliers will affect the scores. This led to no meaningful improvement in predictive performance, where RMSE went slightly up to 9.93. Taking this minor change into account, the original cleaned dataset was retained for the final model. Hence, results are stable and not materially influenced by a tiny number of unusual observations.

Conclusion and Recommendations

In conclusion, it is proposed that duration on the company website is the most essential aspect for reaching high order profit at SweetAroma. Customers having longer browsing time tend to shop more, whereas other variables associated with customer background have almost no impact on individual order value.

Considering all these relations, model referred as linear regression was generated to predict revenue of orders and evaluated using RMSE. On the validation set, the model reached an RMSE of £9.89 and a MAE of £7.83. They show that predicted figures are generally within £8–£10 of the real order value. Additionally, despite the occurrence of

large variability, R^2 of 0.42 represents that a meaningful amount of the variation in customer expenditure can be explained.

From a commercial perspective, the goal of the company should be prioritizing strategies that increase customer engagement during website visits. Enhancing product discovery, improving website usability, and personalizing content throughout the browsing experience may be better to reconsider. If such measures are applied, it is likely to hit company targets. Besides, creating richer behavioral data can bring to more accurate estimations in future models.

Prediction of Future Customer Revenue

After evaluation and training steps, the final model applied to the dataset involving 20 new customer orders. The model generated revenue predictions ranging from approximately £17.77 to £67.06 (Table 1).

Table 1. Predicted order-level revenue (£) for new customers.

Order	Prediction
1	63.95
2	62.54
3	47.92
4	55.29
5	46.33
6	58.11
7	64.23
8	56.43
9	59.63
10	64.84
11	60.36
12	47.47
13	52.53
14	17.77
15	59.68
16	67.06
17	44.10
18	51.52
19	62.10
20	64.34

Part 2: Multi-Criteria Decision Analysis for Autonomous Robot Selection

Introduction

AutonomousShipment is a start-up, researching autonomous mini robots for last-mile logistics operations from various stores to customers' doorsteps. However, finding the right robot for deployment is a risky decision-making responsibility, mainly because of balancing multiple conflicting criteria. Therefore, the main goal of this analysis is providing evidence-based decision-making for last-mile logistics operations.

The main problems that must be considered simultaneously in this situation are operational performance, cost efficiency, reliability together with factors related to designs of the robots. Despite the fact that automation promises faster deliveries and lower operational costs, none of the seven generated models (Alpha, Bravo, Charlie, Delta, Echo, Foxtrot, and Gamma) outperforms others across all characteristics.

The following body of the report covers two business questions:

1. Which robot prototype best supports large-scale delivery operations according to management priority criteria?
2. Which robot should be chosen when taken technology value into account such as battery capacity, cost efficiency, and reliability?

To answer these questions, Multi-Criteria Decision Analysis (MCDA) have been applied to examine types of the robot. In this analysis, the technique called Weighted Sum Model (WSM) has been used to rank robots under management priorities, whilst the Analytic Hierarchy Process (AHP) was another method employed to derive criterion weights for the second case.

Methodological Rationale

Robot prototypes can be considered as a finite and predetermined set of alternatives in these scenarios. They were evaluated against a provided set of measurable data; hence the goal here is to identify a total score of every robot by creating ranking list under different strategic options. Taking the mentioned characteristics into account, a multi-attribute MCDA method appeared to be most suitable.

The Weighted Sum Model (WSM) was chosen because of its appropriation to situations in which criteria are easy to quantify and normalize. Moreover, its simplicity and interpretability make this approach widely used in practice. Another advantage of the WSM is that it enables decision-makers in acknowledging how individual criteria assist to the final score in detail.

In the second case, the Analytic Hierarchy Process (AHP) was applied with the goal of acquiring criterion weights prioritizing features that are technology-focused. In this technique, thanks to pairwise comparisons strategic preferences can be ranked into quantifiable weights. After combining them with WSM, an evolution that is both apparent and consistent with strategy can be derived.

Data and Criteria Overview

Seven robot models (Alpha, Bravo, Charlie, Delta, Echo, Foxtrot, and Gamma) have been weighted considering seven criteria: carrying capacity, battery size, speed, mobility, aesthetic, cost per unit, and reliability. The differences between robot types were observed in the raw performance measurements but it was hard to select the winner from the raw measurements.

To make it clear, a bar chart of cost per unit about all robots illustrates one of the most influential criteria (Figure 8. Cost per unit by robot prototype (lower is better).). From the graph, it can be seen that Bravo and Gamma are on the top of list for the most cost-efficient types (around 4,500–5,000), with Alpha ranked first at 10,000. Specifically in terms of cost-sensitive strategies, such gaps will have a great impact on later rankings.

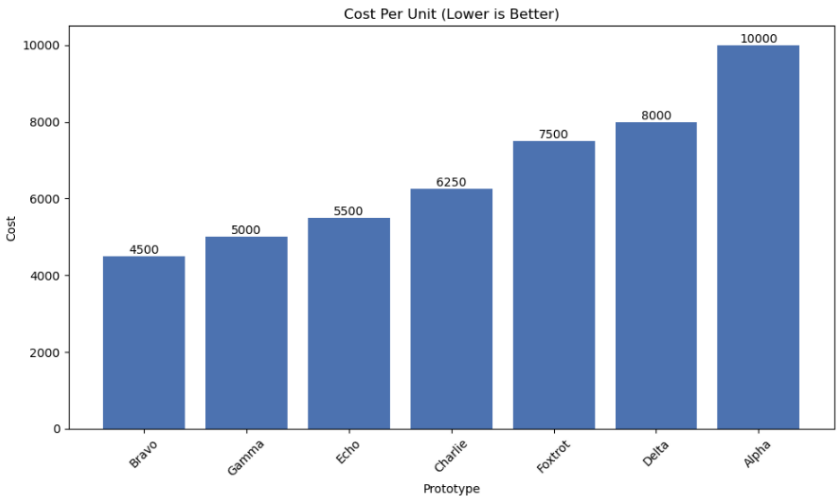

Figure 8. Cost per unit by robot prototype (lower is better).

General visualizations of the raw data shows that no robot performs utmost across all criteria, which makes sure that there is a requirement for a structured MCDA methodology **(**Figure 9 **and** Figure 10**)**.

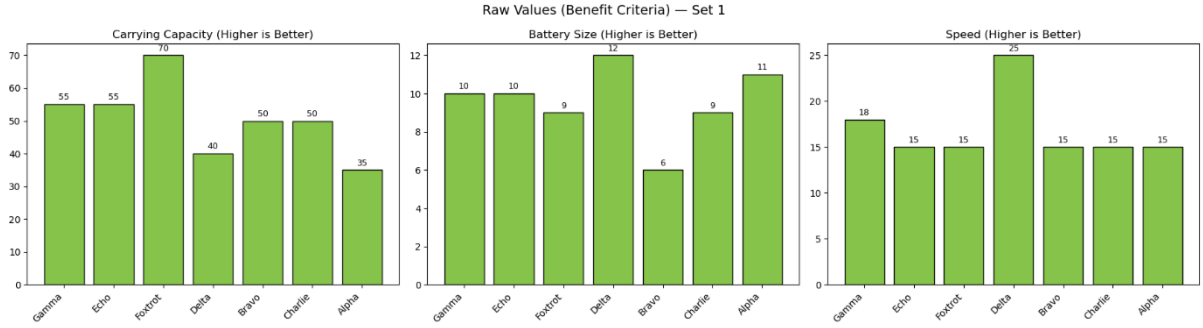

Figure 9. Raw Values (Benefit Criteria) showing carrying capacity, battery size, and speed across robot prototypes.

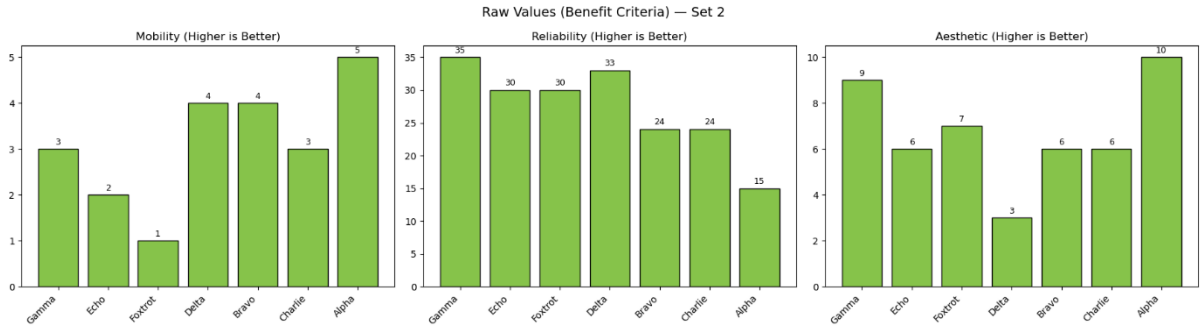

Figure 10. Raw Values (Benefit Criteria) showing mobility, reliability, and aesthetic performance across robot prototypes.

Normalization of Criteria

In this section, min–max normalization was employed to normalize all values between 0 and 1. This is because the criteria are measured on different scales and include both benefit and cost dimensions. Therefore, benefit criteria should be scaled to get trustworthy performance, while cost per unit should be inversely scaled since lower costs means higher normalized scores.

Figure 11 is a visual for the normalized criteria considering the management ranking. Darker cells here show stronger relative performance. It also demonstrates that Gamma performs well on key features such as cost efficiency and reliability, while the figures for the robot called Alpha are quite low in several aspects. This is the evidence that the normalization process helps to preserve relative differences by enabling effective aggregation using the WSM.

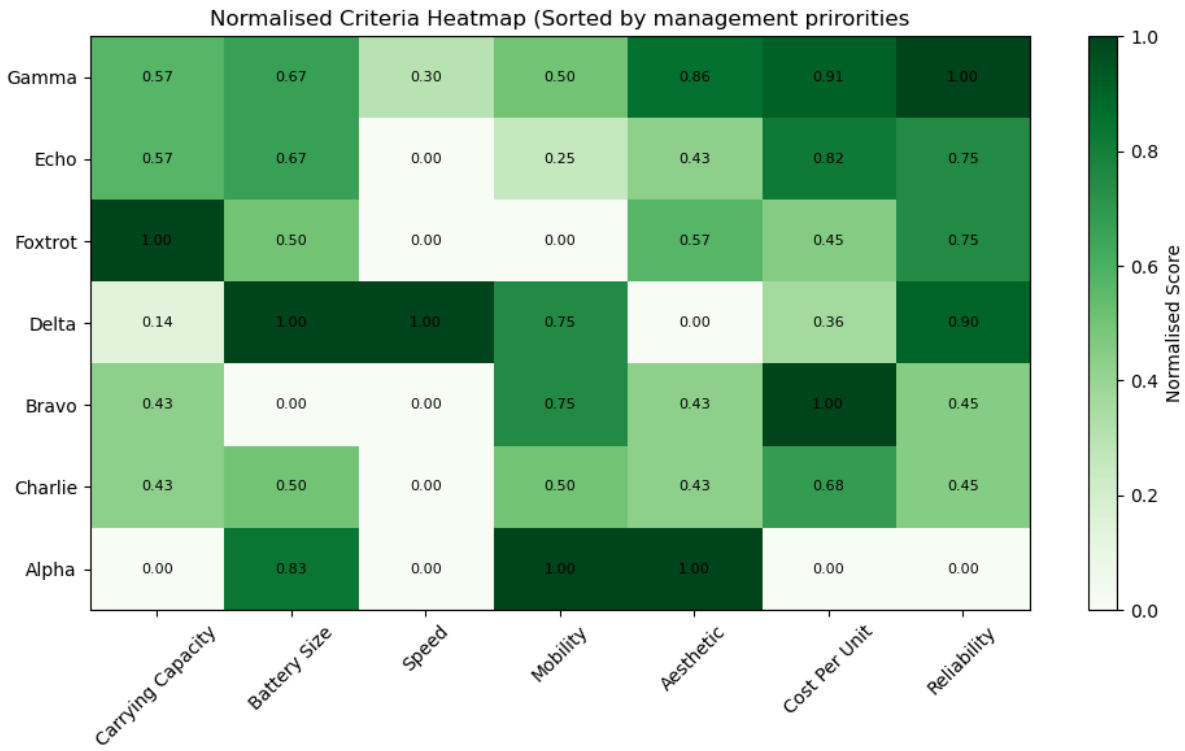

Figure 11. Normalized criteria heatmap (sorted by management priorities).

Question 1: Management-Oriented Operational Strategy

Tas mentioned before, the first task focuses on operating at scale and maximizing large shipment deliveries. Considering management inputs, carrying capacity (0.28) and cost per unit (0.24) achieved the highest values, followed by battery size and reliability, although aesthetic accounted for the lowest in importance. Applying the WSM resulted in the rankings shown in Table 2. It can be seen that Gamma takes the first place at 0.70, followed by Echo at 0.59 and Foxtrot at 0.58, while Alpha ranks last with the minimum of 0.25.

Table 2. Criterion weights and highest-scoring robot prototypes with corresponding scores and ranks.

Criterion	Weight	Prototype	Score	Rank
Carrying Capacity	0.28	Gamma	0.7031	1
Cost Per Unit	0.24	Echo	0.5902	2
Battery Size	0.16	Foxtrot	0.5819	3
Reliability	0.12	Delta	0.5353	4

Speed	0.08	Bravo	0.4911	5
Mobility	0.08	Charlie	0.4748	6
Aesthetic	0.04	Alpha	0.2533	7

Figure 12 visualize the stacked WSM contributions, where strong cost efficiency, reliability, and carrying capacity are the essential reasons of top position Gamma. Besides that, Foxtrot has the top carrying capacity but its higher cost and reduced mobility drop its final score.

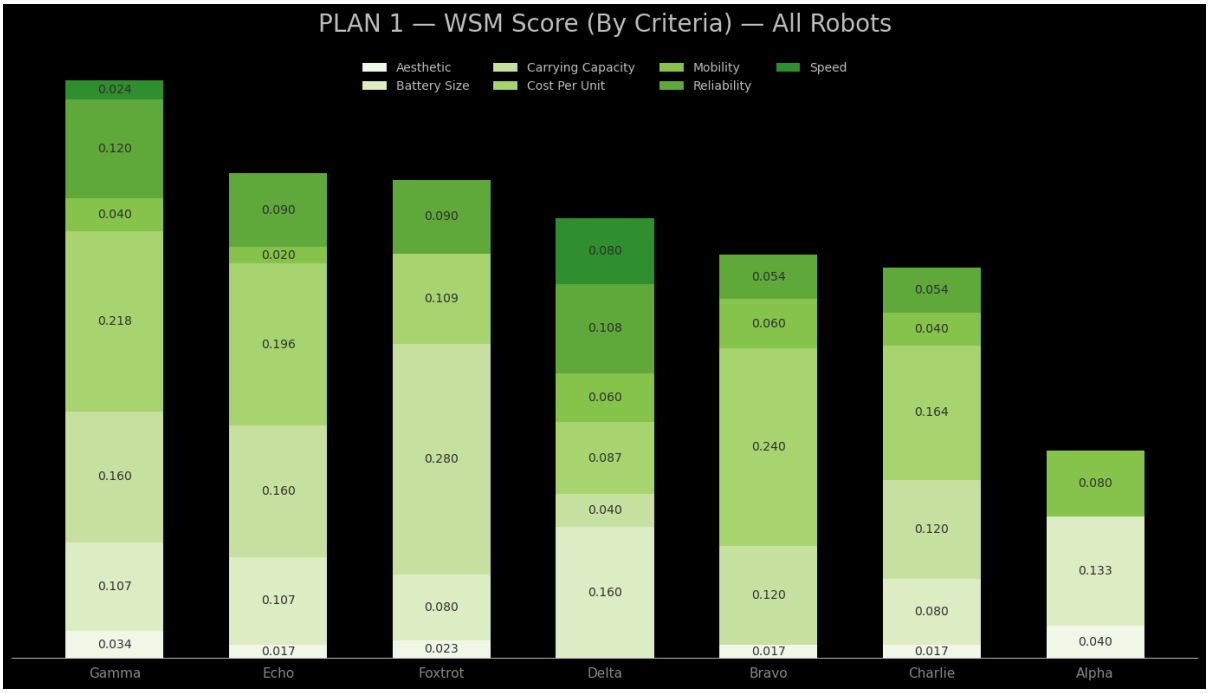

Figure 12. WSM scores by criterion for all robot prototypes based on management priorities.

Question 2: Technology-Focused Strategy Using AHP

The second business problem is related to a technology-selling strategy. To make it clear, battery size, cost efficiency, and reliability are defined to be the main value contributors. Criterion weights were calculated with the Analytic Hierarchy Process (AHP) approach.

Table 3. Plan 2 –WSM results by criterion, showing the highest-scoring prototype and ranking.

Criterion	Weight	Prototype	Score	Rank
Battery Size	0.3001	Gamma	0.7809	1
Cost Per Unit	0.3001	Delta	0.6811	2

Reliability	0.2049	Echo	0.6691	3
Carrying Capacity	0.0863	Foxtrot	0.5395	4
Speed	0.0429	Charlie	0.5151	5
Mobility	0.0429	Bravo	0.4712	6
Aesthetic	0.0227	Alpha	0.3157	7

In this case, battery size and cost per unit gained the importance, both receiving roughly 30% weight, followed by reliability just over 20% as required. The above rankings (Table 3) reveal Gamma still stays first (0.7809), with Delta rising to second place (0.6811). As strong battery size and reliability are heavily weighted under this business plan, they led to the improvement of score for Delta. These shifts ere apparently visible in Figure 13, where measurements stemming from technology-related criteria dominate the overall scores.

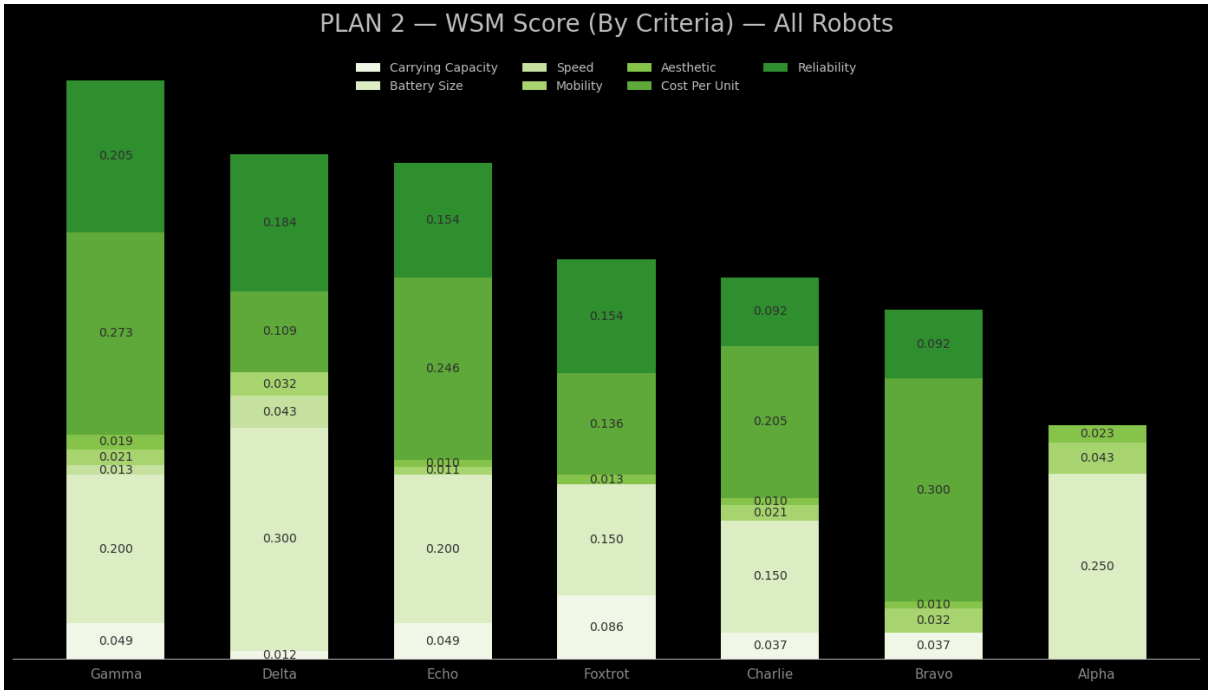

Figure 13. Plan 2 –WSM scores by criterion across all robot prototypes.

Comparison and Selection of the models

The assessment rates have been used to compare the two strategic plans. Based on the results found, Gamma is recommended for the first plan considering the fact that balanced performance of Gamma model make it rank first. As mentioned above, its scores become considerable (0.70 and 0.78), which is demonstration of stability under different weighting conditions.

There is a small difference between competitors in the second plan, Delta and Echo models are separated by only 0.012. This points out that the ranking cannot be defined solely using the sum of the scores. That is where additional business-relevant criteria must be taken into account. Since factors related to battery capacity and reliability are the main value derivers, the robot called Delta should be preferred as the second robot, which dominates Echo on both criteria.

Conclusion and Recommendations

To sum up, this part of report covers the application of a structured WSM, supported by normalization and AHP-derived weights. As it was requested to recommend two prototypes, one for each of two business plans, evaluation of robot types and the selection of models for a trial phase were performed. The application of the WSM method led to a clear comparison of alternatives under different strategic plans.

Taking the results into account, **Gamma** is recommended under the operational strategy (Plan 1) due to its noticeably strong overall performance whereas **Delta** is proposed under the technology-focused strategy (Plan 2). Moreover, Delta robot performs well specifically on battery capacity and reliability, and these are central to the technology objective. Thus, choosing these proposed robots lets the company consider both operational and technological aspects, and recommendations are solidified by Gamma's stable performance for both plans.

Appendices

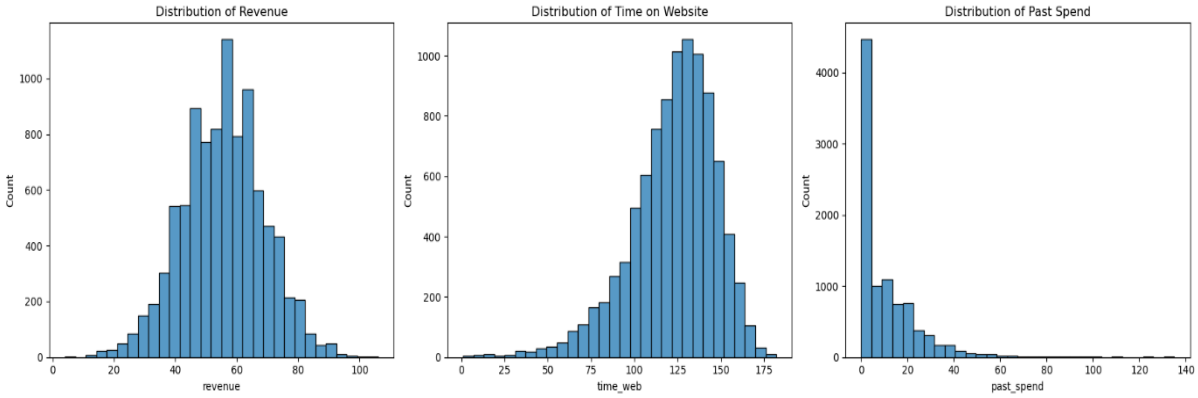

Figure 14. Distributions of order revenue, time spent on the website, and past customer spending.

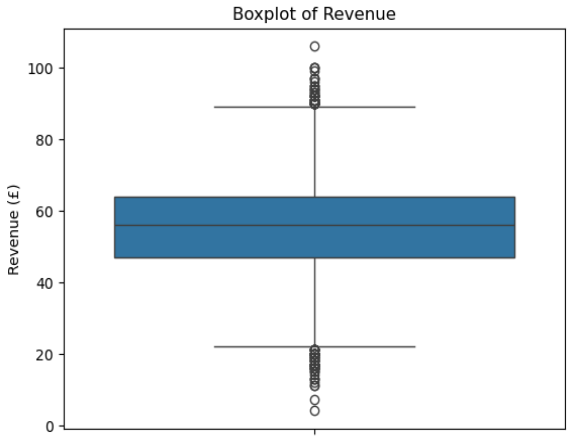

Figure 16. Boxplot of order revenue

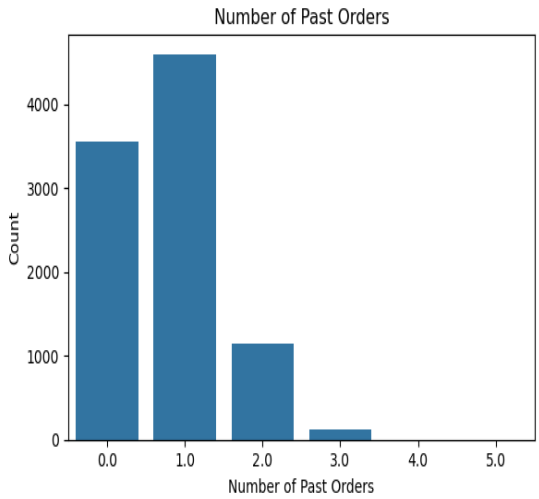

Figure 15. Frequency of customers by number of past orders

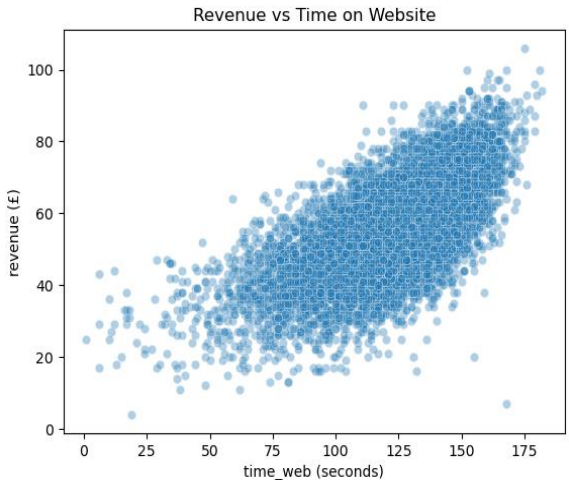

Figure 17. Relationship between order revenue and time spent on the website.

Figure 18. Relationship between order revenue and past customer spending.